# Arista Access Network Node KR-ACC-56 QoS Congestion Event ALARM_6982 Resolution SOP

**Node:** KR-ACC-56 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_6982

## Description

The Access Network Node KR-ACC-56, operating under the Arista vendor CLI dialect, is prone to QoS congestion events, specifically microbursts and interface tail drops, due to QoS queue starvation. This phenomenon occurs when the node's forwarding plane is overwhelmed by bursty traffic, leading to packet loss and subsequent queue drops. The underlying hardware mechanics, such as the use of egress queues and buffer management algorithms, exacerbate this issue. Furthermore, the Arista switch's reliance on software-based QoS policing and shaping mechanisms can introduce latency and jitter, contributing to the occurrence of microbursts. As a result, it is essential to investigate and rectify this issue promptly to prevent further network degradation.

## Pre-requisite Checks

- Verify that the KR Puram Access site is accessible and all necessary equipment is powered on.

- Confirm that the redundant link status is operational and functioning correctly.

- Ensure that all necessary personnel, including remote hands, are dispatched to the site to facilitate troubleshooting and repair efforts.

## Potential Root Causes

| Cause | Probability |
|---|---|
| QoS Queue Starvation | High |
| Interface Tail Drops | Medium |
| Microbursts | Low |

## Diagnostic Commands & Expected Outputs

```
Command: show queue statistics, Expected Output: Queue statistics for the affected interface, including packet loss and drop counts., Description: This command will provide valuable insights into the queue statistics for the affected interface, helping to identify the root cause of the QoS congestion event.

Command: show interface statistics, Expected Output: Interface statistics, including errors and discards., Description: This command will provide a comprehensive overview of the interface statistics, including errors and discards, which can help isolate the issue.

Command: show qos policy statistics, Expected Output: QoS policy statistics, including policy hits and misses., Description: This command will provide information on the QoS policy statistics,
```

```
including policy hits and misses, which can help diagnose QoS-related
issues.
```

## Resolution Steps

Step 1: Step: Verify and Adjust QoS Policies, Command: configure qos policy police cir 100000000 bps, Description: Verify the existing QoS policies and adjust them as necessary to ensure that the policing and shaping mechanisms are configured correctly to prevent QoS queue starvation.

Step 2: Step: Verify and Adjust Interface Configuration, Command: configure interface no buffer size 10000, Description: Verify the existing interface configuration and adjust it as necessary to ensure that the buffer size is not too small, leading to interface tail drops.

Step 3: Step: Verify and Adjust Egress Queue Configuration, Command: configure egress queue max-size 100000, Description: Verify the existing egress queue configuration and adjust it as necessary to ensure that the maximum queue size is sufficient to prevent microbursts.

## Post-Check Verification

```
Command: show queue statistics, Expected Output: Queue statistics for
the affected interface, including packet loss and drop counts.,
Description: Verify that the queue statistics have improved after
implementing the resolution steps.

Command: show interface statistics, Expected Output: Interface
statistics, including errors and discards., Description: Verify that
the interface statistics have improved after implementing the
resolution steps.

Command: show qos policy statistics, Expected Output: QoS policy
statistics, including policy hits and misses., Description: Verify
that the QoS policy statistics have improved after implementing the
resolution steps.
```

## Roll-back Procedures

- Step: Restore Original QoS Policies, Command: configure qos policy no police cir 100000000 bps, Description: Restore the original QoS policies if the fix fails.

- Step: Restore Original Interface Configuration, Command: configure interface buffer size 10000, Description: Restore the original interface configuration if the fix fails.

- Step: Restore Original Egress Queue Configuration, Command: configure egress queue max-size 10000, Description: Restore the original egress queue configuration if the fix fails.